Network Threat Hunting and Incident Investigation

Using Wireshark

Internship Task 3

Money Mitra Network

Project Type	Offline PCAP Analysis and Threat Investigation
Analysis Period	19-20 October 2026
Analysis Environment	Kali Linux VM; Wireshark
Dataset	2024-10-23-Redline-Stealer-infection-traffic.pcap
Submission Date	20 October 2026

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October 2026

1. Executive Summary

This investigation examined a publicly available network packet capture (PCAP) associated with a RedLine Stealer infection traffic exercise dated 23 October 2024. The analysis was performed offline using Wireshark in Kali Linux to identify suspicious network communications, extract indicators of compromise (IOCs), correlate packet-level evidence, and develop defensive detection recommendations.

The capture contained 8,807 packets over approximately 25.68 seconds. Analysis identified HTTP SOAP requests using the actions Endpoint/CheckConnect and Endpoint/GetUpdates to the external endpoint 188.190.10.10:55123. The GetUpdates request contained structured host information. DNS and TLS activity involving api.ip.sb and 104.26.13.31 was also observed.

These findings warrant further investigation in the context of the supplied training capture. However, the network evidence alone does not independently establish successful credential theft, data exfiltration, persistence, or the full impact of a real-world compromise.

2. Scope and Objectives

2.1 Scope

The investigation was limited to offline analysis of the supplied PCAP file. No live scanning, malware execution, or direct contact with the observed external endpoints was performed.

2.2 Objectives

- Examine network protocols and communication patterns using Wireshark.
- Identify potentially suspicious endpoints, ports, hostnames, and application-layer requests.
- Correlate relevant packets and timestamps to construct an evidence-based timeline.
- Record observed indicators of compromise and distinguish observations from inferences.
- Develop practical display filters, defensive detection logic, and incident response recommendations.

2.3 Evidence Source

Dataset: `2024-10-23-Redline-Stealer-infection-traffic.pcap`
SHA-256: `6498af315e9206c6ffc1ed5a488b28d1230c2fb44f2252bb5f05c3f8f68ce311`

3. Tools and Environment

The following tools and environment were used during the investigation:

- Operating System: Kali Linux
- Packet Analysis: Wireshark 4.6.6
- PCAP Metadata Inspection: capinfos
- Integrity Verification: sha256sum
- Documentation: Mousepad text editor and Google Docs
- Supporting Analysis: Command-line utilities available in Kali Linux

4. Investigation Methodology

4.1 PCAP Validation and Integrity

The supplied PCAP was inspected to establish its packet count, capture duration, timestamp range, and integrity. The capture contained 8,807 packets, occupied approximately 6,205 kB, and covered approximately 25.68 seconds.

The SHA-256 hash was calculated to support evidence integrity verification: 6498af315e9206c6ffc1ed5a488b28d1230c2fb44f2252bb5f05c3f8f68ce311

4.2 Protocol and Conversation Analysis

Wireshark's Protocol Hierarchy and Conversations statistics were used to understand the traffic composition and identify significant communication endpoints. TCP conversations were reviewed to locate relevant exchanges for further inspection.

4.3 Application-Layer Inspection

HTTP requests and responses were examined to identify SOAP actions, request destinations, response codes, and relevant application-layer content. Particular attention was given to Endpoint/CheckConnect and Endpoint/GetUpdates.

4.4 DNS and TLS Correlation

DNS queries and responses were examined and correlated with TLS activity to identify the hostname api.ip.sb and the associated destination address 104.26.13.31. The observed DNS and TLS traffic was treated as contextual evidence rather than proof of malicious activity.

4.5 Evidence Recording and IOC Extraction

Relevant packet numbers, timestamps, endpoints, ports, and application-layer indicators were recorded in the investigation notes and IOC register. Findings were classified according to whether they were directly observed or inferred from context.

4.6 Limitations

The investigation was based solely on the supplied packet capture. It did not include endpoint forensics, malware execution, process analysis, or corroboration from host logs. Encrypted traffic content could not be fully assessed without the necessary decryption material. Conclusions are therefore limited to the evidence visible in the capture.

5. Network Analysis and Findings

5.1 Overview of Network Traffic

The packet capture contained 8,807 packets exchanged over approximately 25.68 seconds. Protocol Hierarchy and TCP Conversations statistics were used to understand the traffic composition and identify conversations requiring closer inspection.

The analysis focused on HTTP application-layer exchanges, DNS resolution, and TLS connection metadata. The findings below describe observed packet evidence and distinguish it from conclusions that would require additional investigation.

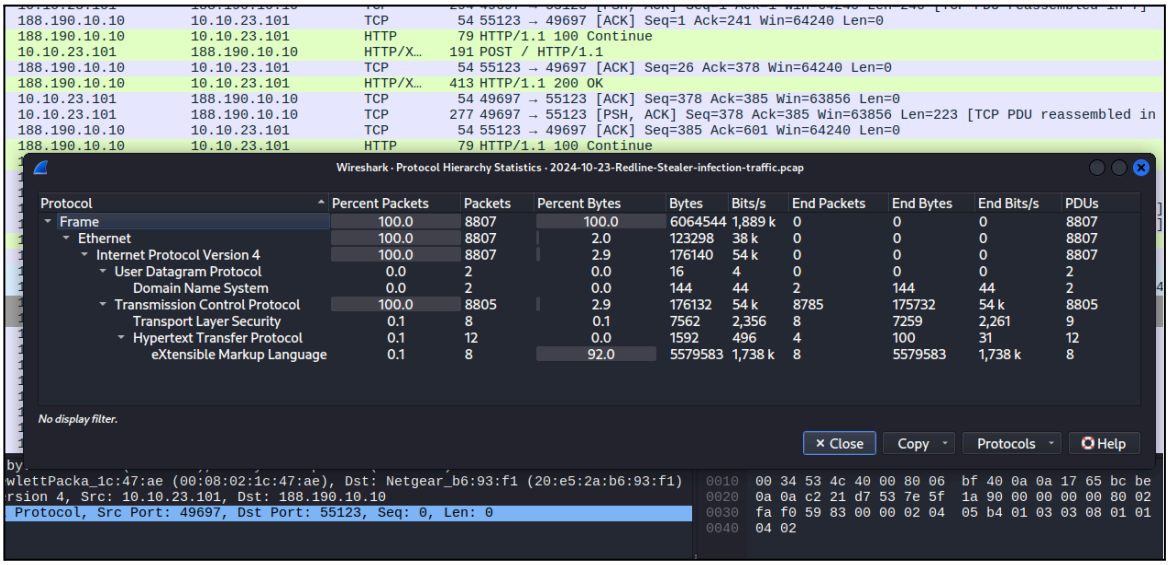


Figure 1. Wireshark Protocol Hierarchy of the Analysed Network Traffic

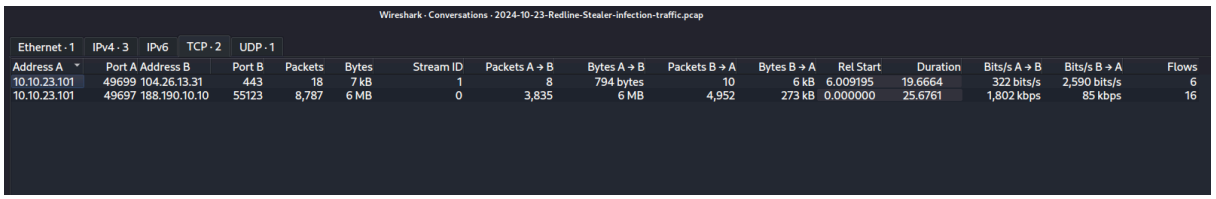


Figure 2. TCP Conversations Identified in the PCAP Capture

5.2 HTTP Check-In Communication

Packet 7, captured approximately 0.343 seconds after the start of the capture, contained an HTTP POST request from 10.10.23.101:49697 to the external endpoint 188.190.10.10:55123. The request used the SOAP action Endpoint/CheckConnect.

Packet 9, approximately 0.502 seconds after capture start, contained the corresponding HTTP response with status 200 OK and a result indicating true.

This exchange demonstrates that the request received an application-level response. The request and response alone do not establish the complete purpose of the communication or prove malicious activity.

5.3 GetUpdates Request and Host Information

Packet 8786, approximately 24.737 seconds after capture start, contained an HTTP POST request using the SOAP action Endpoint/GetUpdates, directed to 188.190.10.10:55123.

The request contained structured host information. Packet 8805, approximately 25.651 seconds after capture start, contained a corresponding HTTP 200 OK response.

The request content and response are relevant to threat hunting because they show information being sent to an external endpoint and a successful HTTP exchange. However, the available packet evidence does not independently prove credential theft, successful data exfiltration, or the full extent of any compromise.

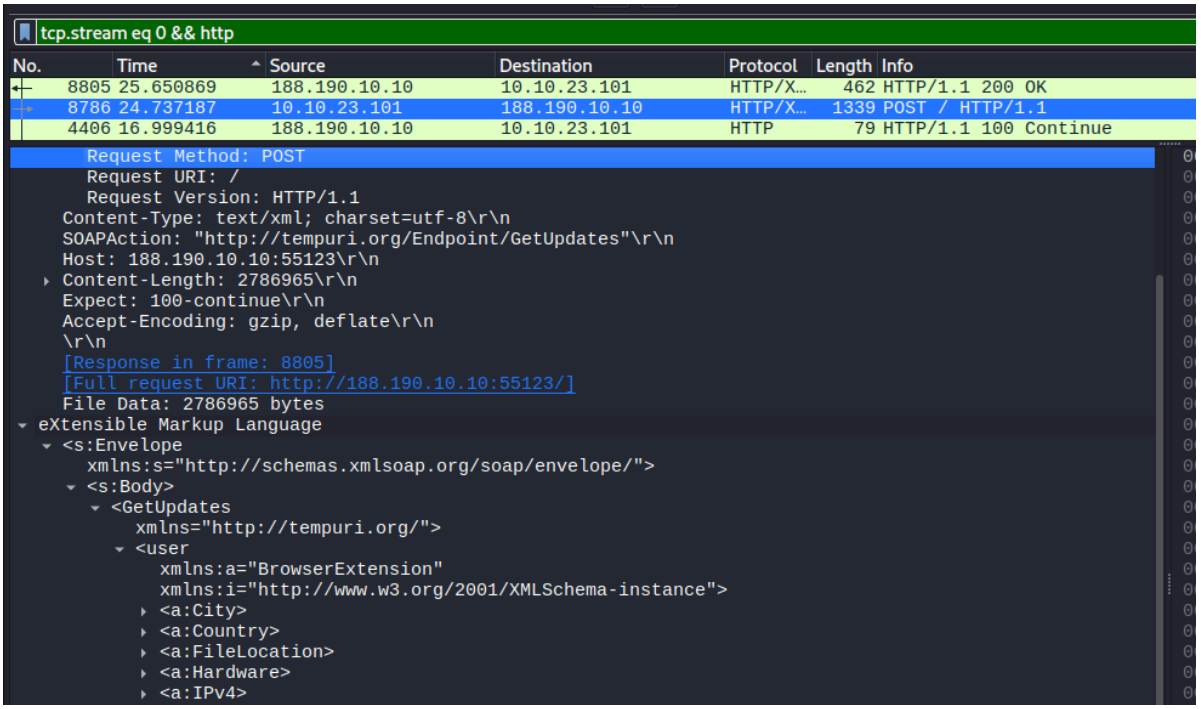


Figure 3. Wireshark Packet Inspection of the HTTP GetUpdates Request

5.4 DNS and TLS Activity

Packets 21–22 recorded DNS activity involving api.ip.sb. The response included a CNAME associated with api.ip.sb.cdn.cloudflare.net and the address 104.26.13.31. TLS activity was also observed with api.ip.sb as the Server Name Indication (SNI).

These observations establish that the hostname was resolved and associated TLS activity occurred. The address is associated with shared Cloudflare infrastructure, so the IP address alone is not sufficient to classify the destination as malicious.

5.5 Preliminary Assessment

The CheckConnect and GetUpdates exchanges warrant further investigation in the context of the supplied RedLine Stealer training capture. The findings should be correlated with endpoint telemetry, process execution records, security logs, and other independent evidence before drawing conclusions about a real-world incident.

6. Indicators of Compromise (IOCs)

The indicators identified during the investigation are summarised in Table 1. The register records each indicator, its supporting packet evidence, confidence, and assessment. The original CSV register is retained as a supporting evidence file.

6.1 Key Indicators

Indicator	Value	Evidence	Confidence	Assessment
IPv4	188.190.10.10	Packets 7, 9, 8786	High	Main remote endpoint; role inferred from context
TCP Port	55123	Main TCP conversation	High	Observed port; not inherently malicious
HTTP SOAP Action	Endpoint/CheckConnect	Packet 7	High	Observed application check-in
HTTP SOAP Action	Endpoint/GetUpdates	Packet 8786	High	Observed host information request
Hostname	api.ip.sb	Packets 21-22 and TLS SNI	High	Contacted; maliciousness unconfirmed
IPv4	104.26.13.31	DNS response and TLS conversation	High	Contacted; shared Cloudflare address; maliciousness unconfirmed

The observed endpoint and application-layer actions are relevant to the investigation. However, an observed IP address, port, hostname, or HTTP action is not inherently malicious. Indicators must be evaluated in their communication context and corroborated with independent evidence where possible.

6.2 Preliminary Severity Assessment

Preliminary Severity: High (Provisional)

The assessment is based on the observed CheckConnect and GetUpdates exchanges and the presence of structured host information in the request. These observations justify further investigation within the context of the supplied RedLine Stealer training capture.

This is a provisional analytical assessment, not confirmation of a live security incident. The packet capture alone does not establish credential theft, successful data exfiltration, persistence, or the full impact on an endpoint.

6.3 Confidence and Attribution Limitations

Confidence in the existence of the recorded network events is high because the relevant packets are directly observable. Attribution of malicious intent is less certain and depends on contextual interpretation.

The hostname api.ip.sb and address 104.26.13.31 were contacted during the capture, but their maliciousness was not independently confirmed. The latter is associated with shared Cloudflare infrastructure, making IP-only attribution unreliable.

7. Detection Filters and Defensive Analysis

7.1 Wireshark Display Filters

The following display filters can be used to reproduce the main observations and investigate similar traffic patterns.

Purpose	Wireshark Display Filter
Identify HTTP requests	http.request
Filter traffic involving the main endpoint	ip.addr == 188.190.10.10
Identify requests containing relevant SOAP actions	http.request && (http contains "CheckConnect" || http contains "GetUpdates")
Inspect DNS traffic	dns
Inspect TLS handshake traffic	tls.handshake

These filters support investigation and triage. They are not, by themselves, proof that a connection or endpoint is malicious.

7.2 Defensive Detection Logic

Detection Rule 1 — Application-Layer Check-In Activity

Generate an analyst alert when an outbound HTTP POST request contains the SOAP action Endpoint/CheckConnect or Endpoint/GetUpdates and is directed to an external endpoint. Initially assign medium priority and validate the destination, application, and surrounding traffic.

Detection Rule 2 — Unusual Outbound Data Transfer

Generate an alert when an external HTTP request exceeds an organisation-defined request-size threshold and the destination or application is unusual for the originating endpoint. Correlate the event with endpoint telemetry and network context before classifying it as possible data exfiltration.

Both rules are conceptual detection recommendations and require adaptation to the monitoring platform and the organisation's baseline traffic.

8. False-Positive Analysis

8.1 Legitimate HTTP Communication

HTTP POST requests, HTTP 200 responses, and SOAP actions can occur in legitimate applications. These characteristics should not be treated as sufficient evidence of malware without additional contextual indicators.

8.2 DNS and TLS Activity

DNS resolution and TLS handshakes are routine network operations. Contact with api.ip.sb does not independently establish malicious intent, and the shared Cloudflare address 104.26.13.31 should not be blocked solely on the basis of this capture.

8.3 Outbound Request Size

Large HTTP requests can be associated with legitimate application activity, uploads, or telemetry. Request size should be assessed alongside destination reputation, application behaviour, endpoint evidence, and organisational baselines.

9. Incident Response Recommendations

1. Preserve the original packet capture and its SHA-256 hash for evidence integrity.
2. If similar activity is found on a real endpoint, isolate the endpoint according to the organisation's incident response procedures when warranted.
3. Correlate network indicators with endpoint detection telemetry, process execution records, DNS logs, proxy logs, and relevant security events.
4. Investigate whether sensitive information was exposed and determine the scope of any confirmed compromise.
5. Reset credentials or revoke sessions if evidence indicates they may have been compromised.
6. Apply network blocks or other containment measures only after validating the indicators and assessing potential operational impact.
7. Monitor for related activity and document investigation decisions, findings, and recovery actions.

These recommendations are conditional guidance for a suspected real-world incident. The supplied training PCAP alone does not establish that these response actions are necessary for an actual compromised device.

10. Evidence Correlation and Timeline

The following timeline records selected events observed in the packet capture. Timestamps are relative to the beginning of the capture.

Relative Time (seconds)	Packet	Observation
0.343266	7	HTTP POST containing Endpoint/CheckConnect from 10.10.23.101:49697 to 188.190.10.10:55123
0.502328	9	HTTP 200 response to the CheckConnect request; result indicated true
5.954098	21	DNS query for api.ip.sb
6.002437	22	DNS response with a Cloudflare-associated CNAME and address 104.26.13.31
24.737187	8786	HTTP POST containing Endpoint/GetUpdates to 188.190.10.10:55123
25.650869	8805	HTTP 200 response associated with the GetUpdates exchange

The sequence establishes the order of selected network events and supports further investigation of the external HTTP exchanges. The DNS and TLS observations provide additional context but do not independently establish malicious activity.

11. Evidence Integrity

The SHA-256 hash recorded for the analysed PCAP is:
6498af315e9206c6ffc1ed5a488b28d1230c2fb44f2252bb5f05c3f8f68ce311

The hash can be recalculated using sha256sum to check whether the file matches the recorded value. The original capture should be retained unchanged, and analysis should be performed on a working copy where practical.

12. Limitations and Conclusion

This investigation identified HTTP SOAP exchanges, DNS resolution, and TLS activity in a public training capture associated with a RedLine Stealer traffic exercise. The CheckConnect and GetUpdates requests, including the structured host information in the latter, warrant contextual investigation.

The analysis is limited to the network evidence available in the PCAP. It does not independently confirm credential theft, successful data exfiltration, persistence, malware execution, or the impact on a real endpoint. Further conclusions would require corroboration from endpoint telemetry and other relevant evidence.

The results demonstrate a repeatable workflow for packet inspection, indicator extraction, evidence correlation, and development of defensive detection logic.

13. Reproducibility Steps

1. Open 2024-10-23-Redline-Stealer-infection-traffic.pcap in Wireshark.
2. Select Statistics → Protocol Hierarchy to review protocol distribution.
3. Select Statistics → Conversations → TCP to inspect TCP endpoints and ports.
4. Apply tcp.stream eq 0 && http.request to inspect relevant HTTP requests in the identified stream.
5. Inspect packets 7 and 9 for the CheckConnect request and response.
6. Inspect packets 8786 and 8805 for the GetUpdates request and response.
7. Apply the dns filter and correlate packets 21–22 with the observed TLS SNI.
8. Apply the display filters documented in Section 7 to reproduce the targeted inspection.
9. Compare packet numbers, timestamps, and observations with the evidence screenshots and supporting packet dissections.

Appendix A — Wireshark Display Filters

Purpose	Wireshark Display Filter
Identify HTTP requests	http.request
Filter traffic involving the main endpoint	ip.addr == 188.190.10.10
Identify requests containing relevant SOAP actions	http.request && (http contains "CheckConnect" || http contains "GetUpdates")
Inspect DNS traffic	dns
Inspect TLS handshake traffic	tls.handshake

These display filters support the reproduction of key observations in the packet capture. They assist with traffic investigation and triage but do not independently establish malicious activity.

Appendix B — Defensive Detection Rules

Rule ID	Detection Purpose	Detection Logic	Priority	False-Positive Considerations
DR-01	Application-layer check-in	Outbound HTTP POST containing Endpoint/CheckConnect or Endpoint/GetUpdates to an external endpoint	Medium	Legitimate SOAP communication may match
DR-02	Unusual outbound transfer	External HTTP request exceeds an organisation-defined size threshold and has an unusual destination or application	Review required	Legitimate uploads and telemetry may match

These rules are conceptual detection recommendations and should be adapted to the monitoring platform and the organisation's normal traffic patterns. Alerts require analyst validation before classifying activity as malicious.